Abiral Shrestha

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Education

Gettysburg College Gettysburg, PA

Bachelor of Science in Computer Science, Minor in Mathematics *Aug. 2024 – May 2028*

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Discrete Mathematics, Linear Algebra
- **Honors:** Dean's List (Fall 2024–Present), Presidential Scholar

Experience

AI/ML Fellow May 2026 – Present

Break Through Tech *Remote*

- Selected from 3000+ applicant for a selective year long AI/ML Fellowship gaining hands on experience with Neural Networks and core ML techniques
- Built and maintained an end-to-end data processing and machine learning pipeline in Python that cleaned, transformed, and modeled 50K+ records.
- Implemented feature engineering and hyperparameter search workflows using scikit-learn and TensorFlow/Keras, achieving 85% classification accuracy.

Product Designer Intern May 2026 – Present

Develop for Good *Remote*

- Collaborated with designers, engineers, and nonprofit stakeholders to deliver a mobile-first marketplace MVP, translating product requirements into production-ready frontend features using Next.js and REST APIs.
- Communicated technical tradeoffs and design decisions across non-technical nonprofit partners, ensuring alignment on scope, feature prioritization, and engineering constraints.
- Partnered closely with engineering peers to establish cohesive brand guidelines and UI component libraries, streamlining the handoff process and accelerating overall development velocity.

Software Engineer Intern March 2026 – May 2026

Sponsy *Remote*

- Engineered the brand-side dashboard frontend using Next.js and REST APIs, enabling brands to onboard and manage their profiles end-to-end.
- Architected the student-side database schema in PostgreSQL, modeling relationships between user profiles, applications, and scholarship records to support core platform functionality.

Projects

CDN Image Optimizer | *TypeScript, Express.js, AWS S3, CloudFront, Docker, Jest* May 2026 – Jun. 2026

- Built an event-driven image optimization service using TypeScript, Express.js, AWS S3, SNS, and CloudFront, enabling direct S3 uploads via presigned URLs and automatic WebP image generation with Sharp.
- Designed a CDN-backed image delivery pipeline that served signed CloudFront URLs and achieved a 96% cache hit rate, reducing warm-request latency by 2×.
- Containerized the application with Docker and developed comprehensive Jest tests to validate presigned upload flows, image processing pipelines, and AWS service integrations.

Foot-Steps | *React, PostgreSQL, AWS S3, Docker, REST APIs* Jan. 2026 – Feb. 2026

- Developed full-stack travel app with React frontend and Node.js/Express backend, automating photo organization into timeline and map views for 100+ photo trips in under 2 minutes.
- Designed cloud-based media pipeline with AWS S3 storage, Prisma ORM, and automatic EXIF metadata extraction to generate GPS-based map pins without manual tagging.
- Containerized frontend and backend services using Docker to ensure 100% environment consistency across development and production.

Graph Analysis Automation | *Flask, TypeScript, Docker, MySQL* Nov. 2025

Schneider Prize for Technology Innovation 2025

- Built a full-stack React/TypeScript and Python FastAPI platform for real-time FTIR spectral comparison, processing 100+ CSV datasets used by MRG Labs engineers.
- Cut manual analysis time by 70% by automating interactive Chart.js graph overlays and server-side batch exports (Pandas, Matplotlib), backed by a REST API with MySQL session auth.
- Engineered weighted-difference scoring algorithms (RMSE, Pearson, hybrid) that flagged oxidation and contamination trends with 90% accuracy.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL (PostgreSQL), C#, C
Frameworks/Libraries: Express.js, FastAPI, Flask, Redis, BullMQ, Next.js, WebSockets, Jest
Tools & Platforms: AWS (EC2, S3, CloudFront, SNS), Docker, GitHub Actions, Postman, Supabase, Git